



Early Detection of Neurological Decline in ICU Patients

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The Problem Came From a Real ICU

"We lose transplant windows because we recognise brain death too late. By the time we formally confirm it, the organs have already deteriorated."

Dr. Samrath, ICU Physician at Dubai Hospital — this project is our direct response to that problem. Every hour of delay in recognising neurological decline is an hour lost for a potential transplant recipient. We set out to close that gap with machine learning.



What We Are Building



An **ML early warning system** that flags ICU patients experiencing neurological decline before formal clinical recognition — giving transplant coordinators a **12-24 hour head start**.

Neurological Trajectory Only

The model detects the direction of neurological decline — not organ suitability.

Real ICU Data

Trained on MIMIC-IV v3.1 — one of the most comprehensive critical care databases available.

Features — What the Model Sees

1. Neurological (Primary Signal)

- **gcs_min** — lowest GCS recorded during the stay
- **gcs_last** — most recent GCS before discharge
- **gcs_trend** — last GCS minus first GCS (negative = declining)
- **coma_depth** — 15 minus gcs_min (higher = deeper coma)

2. Intervention

- **on_sedation** — received midazolam, propofol, or fentanyl

3. Hemodynamic Stability

- **map_mean** — average mean arterial pressure
- **map_trend** — blood pressure trajectory over stay
- **bp_stability** — inverse of MAP volatility (stable = good organ sign)

4. Organ Context (for doctor handoff)

- **lactate_max** — peak metabolic stress marker
- **ph_mean** — average blood acidity
- **creatinine_last** — terminal kidney function
- **bilirubin_max** — peak liver stress
- **urine_output_total** — total kidney output

5. Severity + Composite

- **los_hours** — ICU length of stay (severity proxy)
- **neuro_body_paradox** — $(15 - \text{gcs_last}) / (\text{lactate_max} + 1)$

High score = severe brain failure, mild organ stress = classic brain death dissociation pattern

Challenges We Faced

1

1. No Complete Label

MIMIC-IV has no dedicated brain death column. **ICD code G93.82** exists but only captured **129 cases** out of 93,414 ICU stays. Reasons for under-coding: - Patients die before paperwork is complete - Brain death documented in notes, not coded - Organ donor records de-identified out
So we had a partial label, not a missing one. The challenge was expanding it reliably.



2. Severe Class Imbalance

362 perfect donors vs 10,000+ negatives — a 1:30 ratio that crushed recall.



3. Data Leakage

`on_invasive_vent` was both a label condition and a feature — circular logic baked into the model.



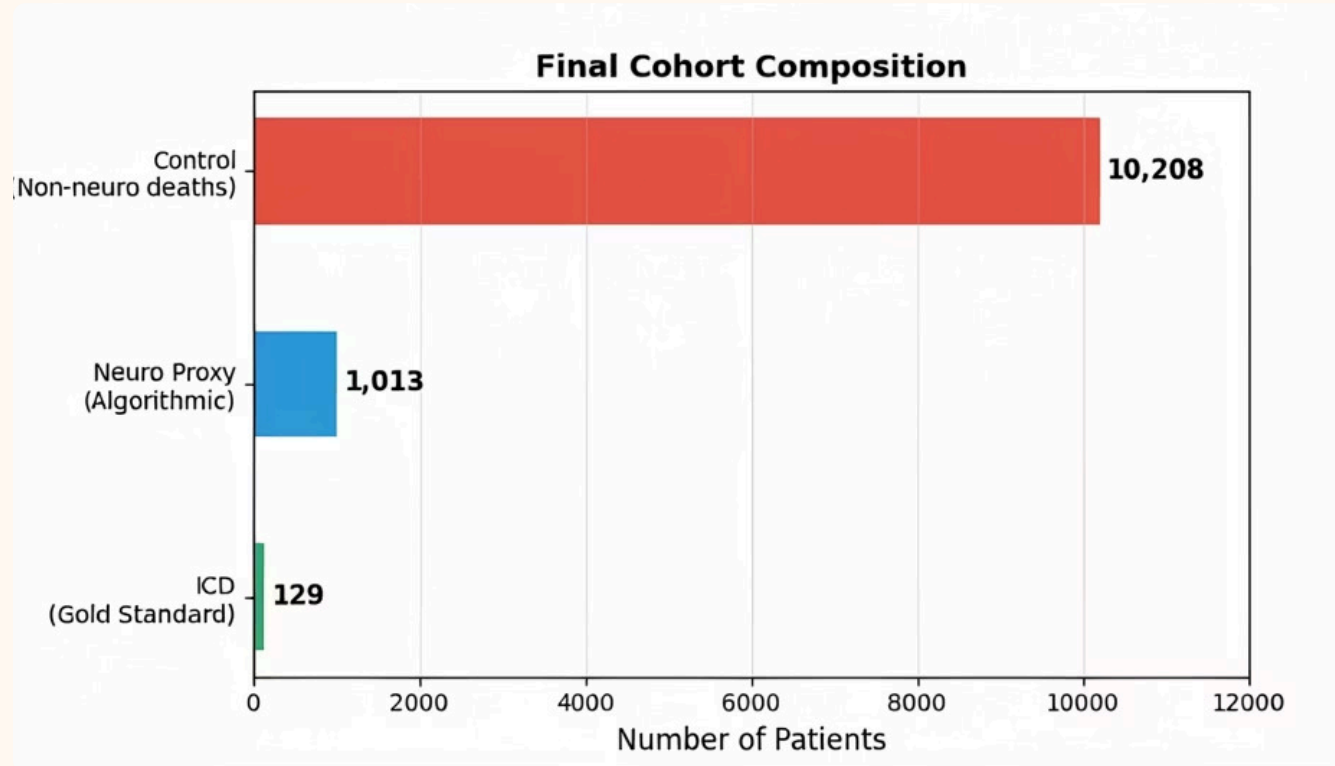
Missing Lab Values

Incomplete data across 11,350 ICU stays required careful imputation strategy. Missing data in ICU is not random — a missing lactate usually means the clinician didn't suspect metabolic crisis. We used median imputation as a conservative baseline while acknowledging that missingness itself carries clinical signal.

Building the Cohort — Who Counts as a Case?

Three Signals, One Cohort(label:0=10208 patients, label:1=1142 patients)

Because no single clean label existed, we triangulated across three independent clinical signals to construct a robust case definition.



1

ICD-10 G93.82(label:1)

129 cases — gold standard, directly coded brain death diagnoses.

2

Clinical Proxy(label:1)

1,013 cases — $GCS \leq 8$ + Invasive Ventilation + In-hospital death.

3

Control Group(label:0)

10,208 cases — died in ICU from non-neurological causes.

Class Imbalance — A Design Problem



Attempt 1: Predict Perfect Organ Donors

Attempt 1 asked: "Is this patient brain dead AND are ALL their organs transplant-ready?" → Very few patients qualify → 362 positives

Positives: **362** | Negatives: **10,000+**
| Ratio: **1:30**

Recall: **50.45%**

clinically useless. The model simply predicted "negative" for everything and was still technically accurate.

The Fix — Make the Label Less Strict:

Redesign asked: "Is this patient experiencing catastrophic neurological decline?"

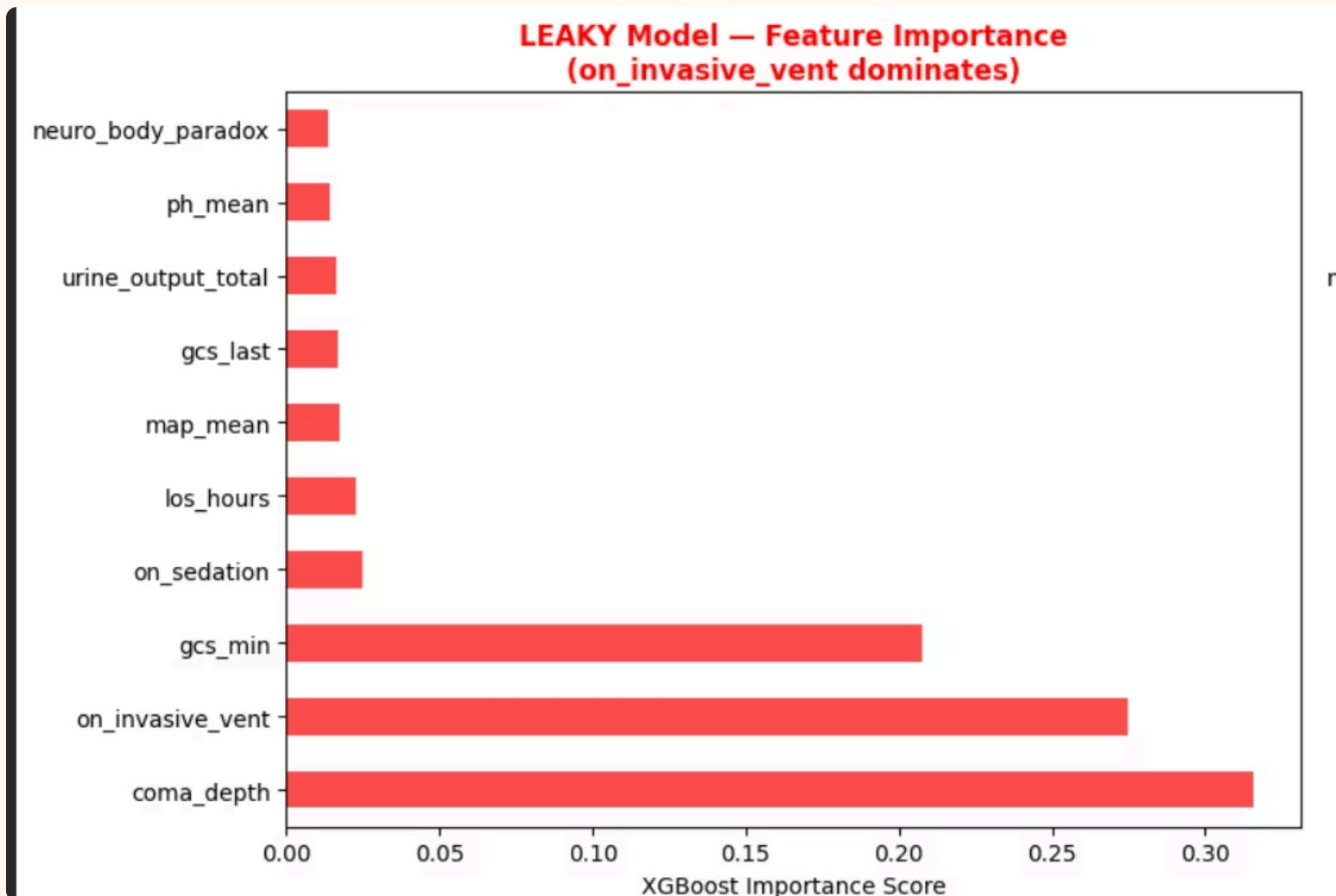
Let doctors assess organs at bedside

How we expanded the positive set:
ICD G93.82 (confirmed brain death) → 129 Clinical Proxy
(GCS≤8 + Vent + Died) → 1,013
Total positives → 1,142

Ratio improved: 1:30 → 1:9 Recall improved: 50% → 87%

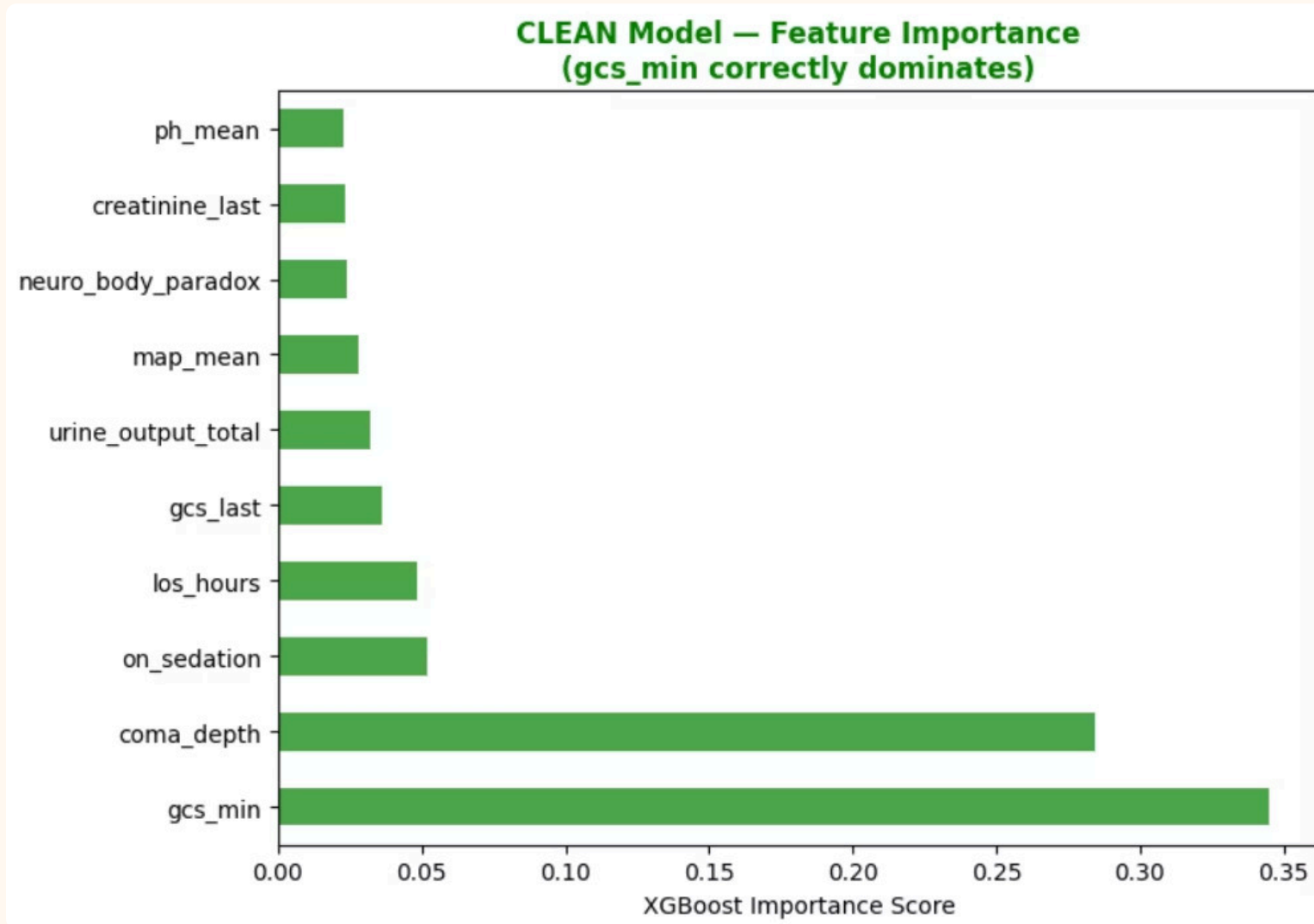
The key insight: organ suitability is a clinical judgement. The ML task is purely neurological trajectory detection.

Correlation is Not Causation — Leakage Detected



Ventilation *follows* brain death — it does not cause it. Including `on_invasive_vent` as a feature meant the model was essentially reading its own label. Removing it cost only **0.018 AUC** while restoring genuine clinical validity. **GCS minimum score** correctly emerged as the dominant predictor after the fix.

After feature processing



After (clean model): gcs_min jumps to #1 coma_depth stays at #2 Both are pure neurological signals

What Changed After Removing on_invasive_vent?

Before (leaky model):

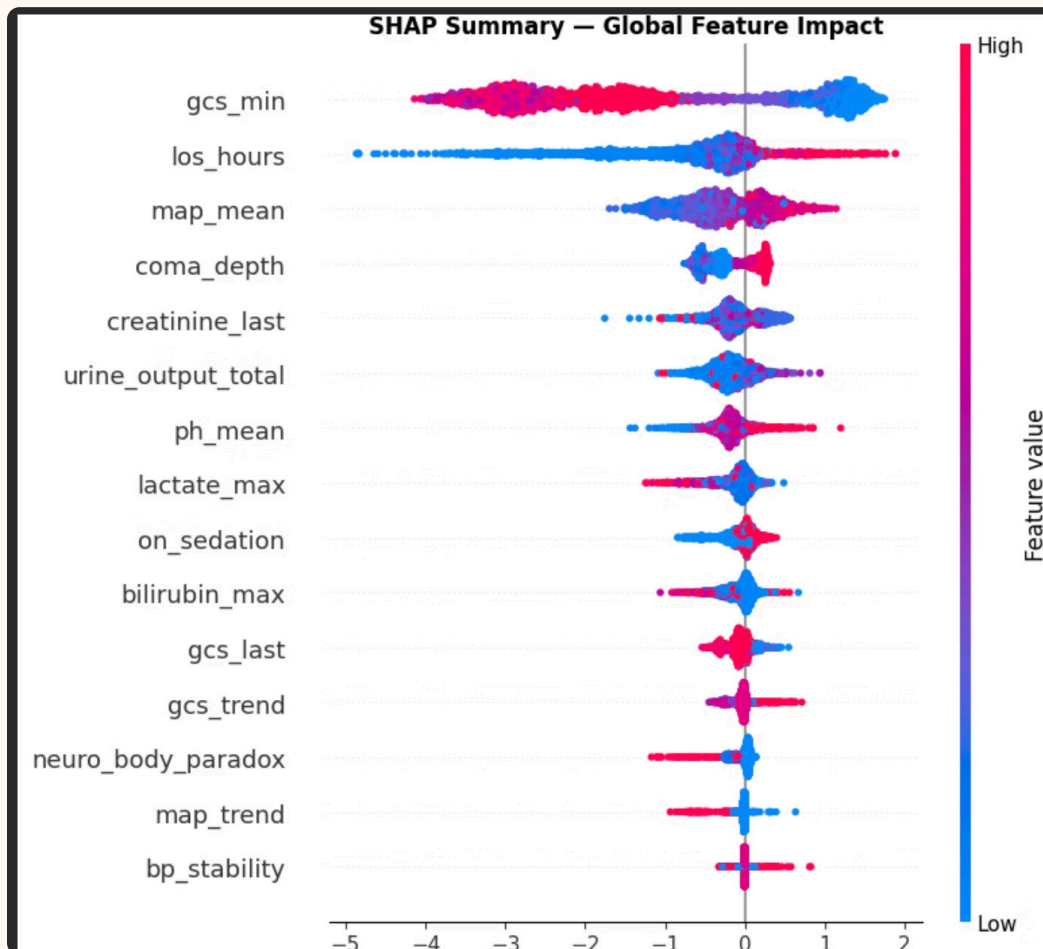
on_invasive_vent was #2 most important feature Model was reading its own label AUC looked great at 0.926 — but it was fake

Why is this the correct result?

Brain death is fundamentally a neurological diagnosis. The dominant predictor should be how low consciousness dropped — not whether a ventilator was used.

A neurologist would tell you exactly this. The model now agrees.

Feature Interpretability (SHAP)



How to Read This Chart

Each dot = one patient Color = feature value Red = high value Blue = low value X position = impact on prediction Right of center = pushes toward brain death Left of center = pushes away from brain death

Reading gcs_min (row1)

Blue dots (low GCS = 3) → far RIGHT → Low GCS strongly flags the patient

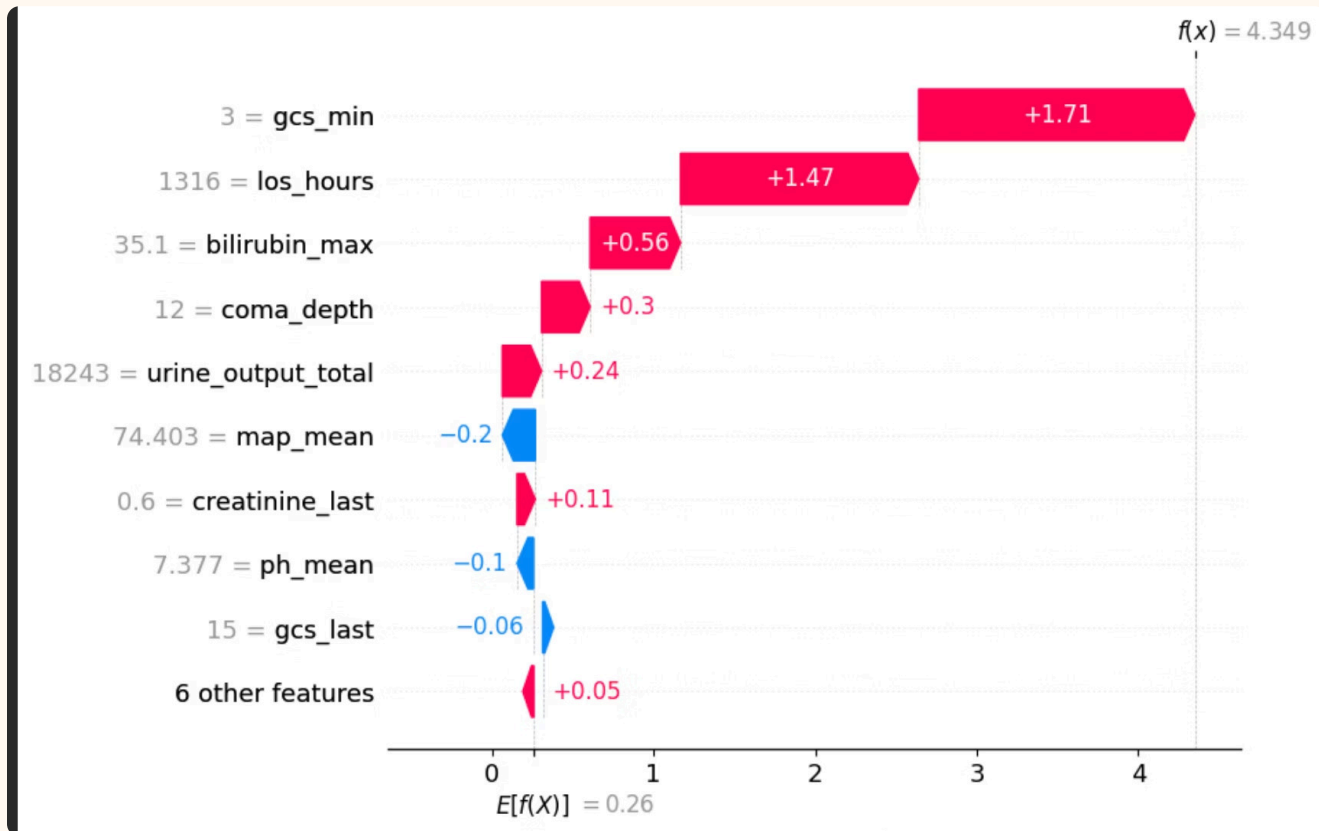
Red dots (high GCS = 15) → far LEFT → High GCS strongly protects against flagging

Clinically perfect. The model learned exactly what neurologists would tell you.

Reading los_hours (row 2)

Blue dots (short stays) → far LEFT → Short ICU stays reduce brain death probability Red dots (long stays) → far RIGHT → Long stays increase probability Confirms los_hours carries real clinical signal — sicker patients stay longer before declining.

Patients example:



This is SHAP's power. For this specific patient — GCS dropped to 3, they spent 54 days in ICU, bilirubin hit 35 — the model gives a risk score of 4.35, far above our threshold of 0.3. A doctor seeing this waterfall immediately understands the reasoning. It's not a black box anymore. This is what clinical AI needs to look like.

Model Training

Threshold	Precision	Recall	F1	Caught	Missed	FalseAlarms
0.2	0.318	0.895	0.470	204	24	437
0.3	0.351	0.873	0.501	199	29	368
0.4	0.376	0.842	0.520	192	36	318
0.5	0.410	0.794	0.541	181	47	260
0.6	0.442	0.706	0.544	161	67	203
0.7	0.488	0.623	0.547	142	86	149

1. Algorithm Choice

Why **XGBoost**? Handles tabular clinical data natively, built-in class imbalance control via `scale_pos_weight`, and produces interpretable feature importances critical for clinical audit.

2. Handling Imbalance

scale_pos_weight = 8.94 Exact ratio of negatives to positives (10,208 / 1,142). This tells the model to penalize missing a positive case **8.94x more** than missing a negative.

3. Train / Test Split

80/20 stratified split **9,080 training** | **2,270 testing Stratified** = same 9% positive rate preserved in both sets. `random_state = 42` for reproducibility.

4. Threshold Selection

Default threshold is **0.5 wrong** for this task. We evaluated **0.2 through 0.7** and selected 0.3. Why 0.3?

Missing a brain death case = lost transplant window. (false negative)

False alarm = one extra clinical review. (false positive)

Recall was prioritized over precision deliberately.

Results and Visualisations

```
... Shape: (11350, 18)
label
0    10208
1     1142
Name: count, dtype: int64

Train: 9080 | Test: 2270
Test positives: 228
scale_pos_weight: 8.94

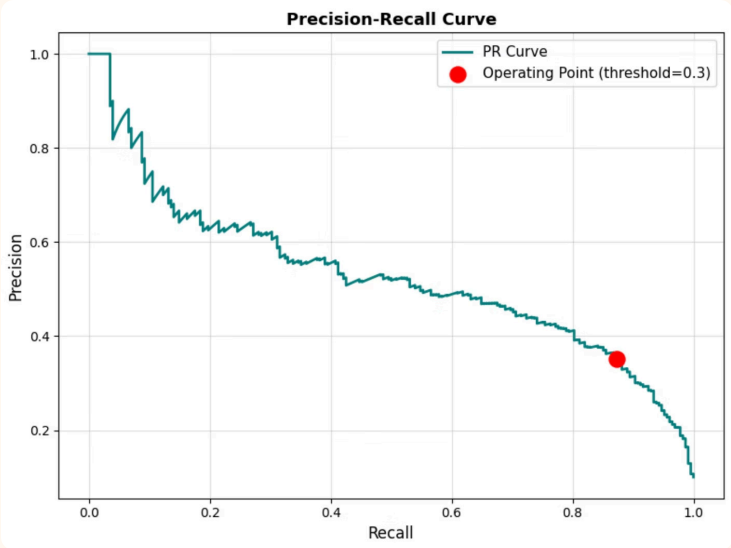
AUC-ROC: 0.9098

--- FINAL PERFORMANCE (threshold = 0.3) ---
              precision    recall  f1-score   support

   Control         0.98         0.82         0.89        2042
  Neuro Decline     0.35         0.87         0.50         228

   accuracy              0.83        2270
  macro avg         0.67         0.85         0.70        2270
 weighted avg         0.92         0.83         0.85        2270

Threshold  Precision    Recall    F1
0.2         0.318         0.895         0.470
0.3         0.351         0.873         0.501
0.4         0.376         0.842         0.520
0.5         0.410         0.794         0.541
0.6         0.442         0.706         0.544
0.7         0.488         0.623         0.547
```



Precision Recall Curve (threshold @0.3)

fig: Result from model training(threshold @0.3)

Recall =87.3%, Precision=35.1%,AUC-ROC=0.91

Confusion Matrix (threshold = 0.3)
199 caught | 29 missed | 368 false alarms

True label	Control	Neuro Decline
	Predicted label	
Control	1674	368
Neuro Decline	29	199

Minimizing false negative

Why is 368 false alarms acceptable? In transplant coordination, the cost of missing a real case (29) far exceeds the cost of a false alarm (368). A false alarm = 10 minutes of review. A missed case = a lost organ donation.

The Key Stat(Recall)

87.3% of all neurological decline cases in the test set were caught. Only 29 patients out of 228 were missed.

Confusion Matrix

Limitation

Every great ML project acknowledges what it *can't* do:

- **Data Source:** MIMIC-IV is from a single center (Beth Israel Deaconess). How would it perform in a rural hospital or a hospital in bihar? (Generalizability).
- **Missing Modalities:** You don't have EEG or Pupil dilation data, which are the gold standards for neurological assessment.
- **Proxy Label Uncertainty** — 1,013 labels are algorithmic, not clinically verified

Our Pipeline



Data Acquisition

MIMIC-IV v3.1 accessed via Google BigQuery.



Label Construction

ICD codes + Clinical Proxy + Control group combined.



Feature Extraction

15 clinical features extracted via SQL queries.



Leakage Detection

Identified and eliminated `on_invasive_vent` circular feature.



XGBoost Model

Threshold = 0.3, `scale_pos_weight` = 8.94 for imbalance correction.



Clinical Alert

87% of neurological decline cases caught and flagged.

Conclusion

We started with a real problem from a real doctor.

Dr. Samrath told us that ICU teams lose transplant windows because brain death is recognized too late. We built a system to change that.

Using MIMIC-IV — real clinical data from 11,350 ICU stays — we trained a model that catches 87% of neurological decline cases with an AUC of 0.91.

But the numbers are not the story.

The story is what we learned building it:

That correlation is not causation — we caught our own leakage and fixed it.

That class imbalance is a design problem — we redefined the question, not just the math.

That missing data is not random — absence of a lab test is itself clinical signal.

That a label is not ground truth — proxy labels carry uncertainty we must acknowledge.

These are not just ML lessons. They are exactly what biomedical informatics demands when the data comes from real patients and the output affects real lives.

The model flags. The doctor decides. That boundary is intentional and important.

We did not solve organ donation. We gave transplant coordinators a head start.